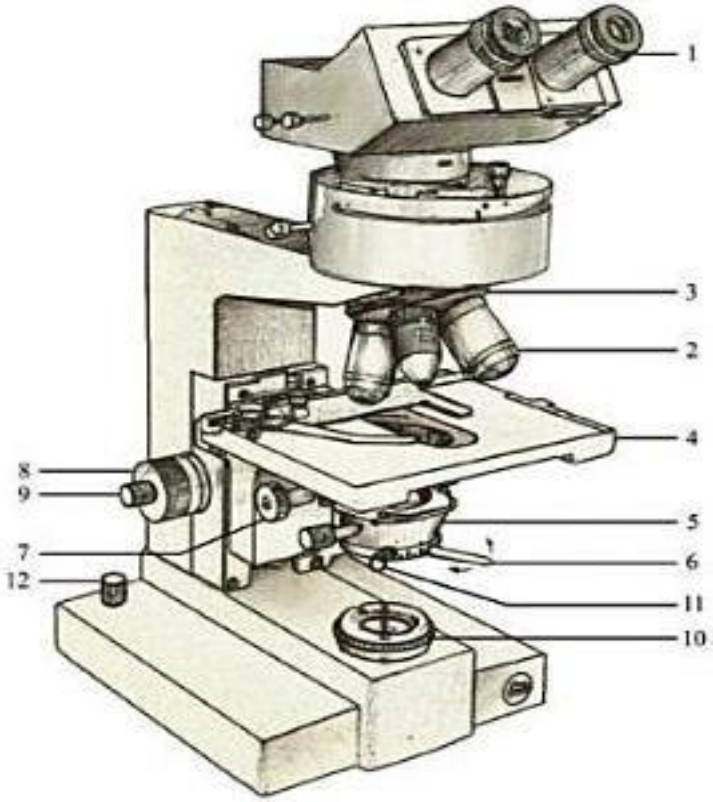
Getting to know a light microscope

Parts of the compound microscope

The following diagram illustrates the parts of a typical binocular compound microscope.

Beginning students should familiarize themselves with the location and functions of the following parts of a microscope.



1. **Eyepieces**: Adjustable focus on one or both to compensate for eyesight differences between individual users.
2. **Objective lenses**: There may be several of these. Usually 4x, 10x and 45x are included as a minimum.
3. **Nosepiece**: The base that supports the objectives. The nosepiece can be rotated to view the specimen through different objective lenses.
4. Microscope stage and the device for holding a slide in place. This device may consist of simple clips or an apparatus that can be moved with the aid of screws below or above the stage.
5. **Condenser**: This is part of the illuminating system consisting of one or more lenses.
6. **Iris** (aperture) **diaphragm** and the lever or ring whereby it can be opened and closed.
7. Condenser adjustment knob.
8. Coarse focus knob that moves the stage through long distances rapidly.
9. Fine focus knob that moves the stage through short distances more slowly.
10. Field diaphragm
11. **Condenser**: mount with sledge guide and centering screws. The light path through the condenser can be aligned using these screws, so that an even field of illumination is achieved.

12. Light intensity control.

It is recommended that you read the appropriate manual before using a microscope.

Precautions:

- When not in use, microscope should be kept covered.
- After using wet mounts, it is important to clean the objective lenses to remove mounting solution that may have been left there inadequately.
- Dirty lenses should be cleared with a drop of lens cleaner on a piece of lens tissue.
- Do not use other types of tissue, as the lens may be scratched.
- Then, polish the lens using a clean, dry portion of lens tissue and a circular motion with light pressure.

Basic instructions:

These instructions are for a binocular microscope. However, if a monocular microscope is being used, omit step8 and 9.

1. Plug in the microscope.
2. Turn on the light.
3. Place the slide, cover glass uppermost, on the microscope stage.
4. Rotate the 4x(or10x) objective lens into place.
5. Adjust the light intensity so that it is comfortable for viewing.
6. While looking at the slide from the side, raise the stage as far as possible with the coarse focus knob. Never raise the stage with the coarse focus knob while looking through the eyepieces.
7. Looking through one eyepiece only, slowly lower the stage with the coarse adjustment knob until the image of the specimen comes into focus.
8. Adjust the interpupillary distance for your eyes. While viewing the specimen with both eyes, push the eyepieces together or pull them apart until one image is seen.
9. Look through each eyepiece in turn and test whether or not it can be focused individually by rotating the focussing ring near the top of the eyepiece while observing the specimen. If only one eyepiece can be focussed, use the other eye and eyepiece to focus on an object within the specimen. Then, without touching the f focussing knobs of the microscope, look the object into focus by turning the eyepiece focussing ring. If both eyepieces can be focussed, this step can be done in any order.
10. To achieve a greater magnification of the specimen, rotate the 10x(40x) objective into place.
11. Use the fine focus knob to adjust the focus if necessary.
12. If a higher magnification is desired, carefully swing the 40x objective into place.
13. Adjust the light intensity for each specimen at each magnification.
14. Adjust the condenser by raising or lowering it using the condenser adjustment knob, until the most even illumination for each specimen and magnification is obtained. Adjusting the condenser will also achieve maximum contrast. The field diagram should be opened so the light just fills the field of view.
15. To remove the slide, rotate the 40x objective do that either the 4x or 10x or no objective is in place. Only then remove the slide from the stage.

Note: Be sure to indicate steps 13 and 14 for each specimen as they are important for obtaining satisfactory images.

PHARMACOGNOSY

SECTION CUTTING

The most common practice used in the study of the plant tissues is the preparation of temporary or permanent histological slides that can be examined with the aid of an optical microscope. In pharmacognosy, organised drugs, both fresh and in air-dried conditions, can be studied conveniently by examining the nature and arrangement of cells and tissues in thin sections cut transversely and/or longitudinally. Under the optical microscope, tissues are examined by transillumination. Since, tissues are examined for transillumination techniques have been developed for obtaining thin and translucent sections.

A student of Pharmacognosy should be familiar with the art and discipline of section cutting.

Necessary material

1. Razor, flat sided for section cutting
2. Scalpel or sharp pen knife
3. Dissecting Needles
4. Pair of small forceps
5. Watch glasses
6. Microscope slides
7. Cover slips ¾ circles
8. Camel – hair brush
9. Clean handkerchief for polishing slides and cover slips
10. Beaker(100ml)
11. Pencil(hard, pointed)
12. White glazed tile
13. Record Book

How to cut sections

- The razor blade and the materials for section cutting must be flooded with water.
- The piece of material should be long enough to be firmly held between the index finger and the thumb of the left hand.
- The index finger should be held down below the surface of the material without coming in the way of the razor.
- Now, take the razor in the right hand and place it at the top of the material at right angles and draw it across the material in such a way as to give the material a drawing cut.
- Special care must to avoid obliquely cut sections, which are likely to be cut if material is not held up straight in level with the left index finger.
- By slightly increasing the pressure with the razor on the finger it is possible to cut a number of sections without the thumb or the material.
- Enough care should be taken to avoid injury to the finger. With the help of fine camel hair brush, sweep the sections off the razor and suspend in water in a watch glass.
- Select a few thin sections and put them in another watch glass containing liquid.

PHARMACOGNOSY

- The razor, after the use, should be dried and stropped immediately in order to preserve the edge of the blade for further use.
- The lamina of a leaf may be too thin to be cut easily as described above. In such cases, the sections should be cut by sandwiching the material between two pieces of pith.
- Elder pith is most suitable, material since it is soft, yet rigid and is easy to cut. Selected pieces of the material, about 2inches in length, should be used.
- Cut a piece out by making a cut down the centre and then another cut through one side to meet it.
- Insert this cut segment in the pith, the diameter of the pith should be more than the width of the material. The block so formed is then used in the usual manner and the sections are cut together.
- Sweep the sections off the razor with a camel hair brush and place them in water in a watch glass.
- The sections of the pith can be identified easily and removed from the watch glass leaving behind the sections of the material. Select few thin sections and put them in another watch glass containing water.

How to make a preparation for microscopic examination

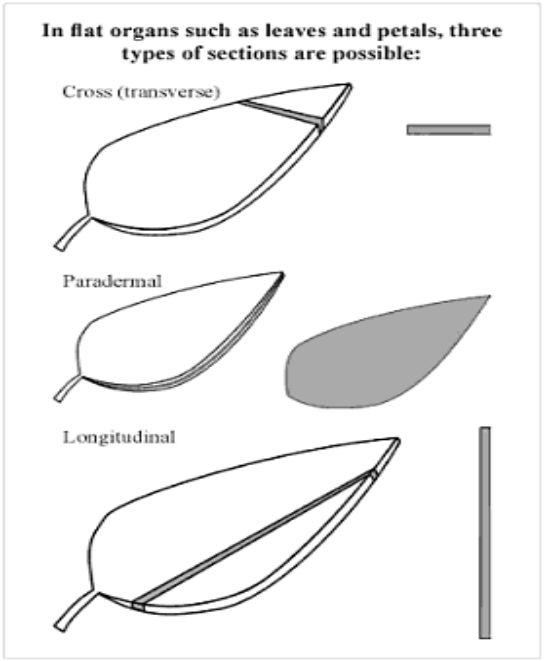
1. Take a clean glass slide and place a drop of glycerine water in the centre of the slide.
2. Hold the cover-slip between in the index finger and the thumb of the left hand and let the edge of cover-slip rest on the side at the left hand edge of the drop.
3. Insert a dissecting needle under the right hand edge of the cover slip and let the latter rest on the needle.
4. Lower the cover-slip slowly on to the drop, taking enough care that the drop of the liquid exactly fills the space between the slide and the cover-slip without any air bubbles being trapped inside. There should be no mountant on the top of the cover slip.
5. Wipe out any excess of mountant on the slide with the help of small pieces of a filter paper.

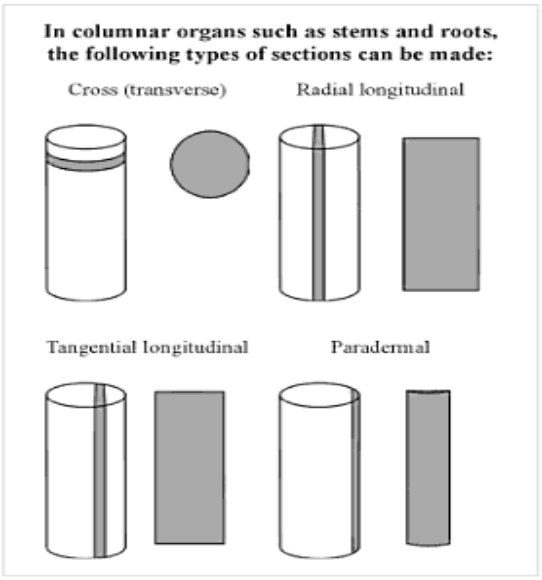
How to focus the material and adjust critical illumination

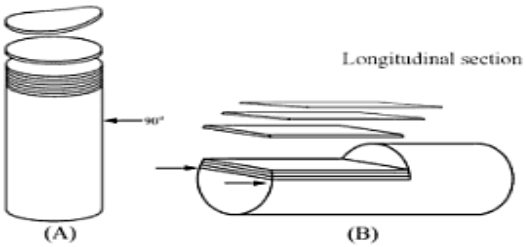
1. Place the slide in position on the stage of microscope, so that the mount lies over the centre of the hold in the stage. Use the spring clips for keeping the slide in place.
2. By means of coarse adjustment, screw down the body tube until the lens of the low power objective comes to within a short distance (about ½") of the cover slip. While lowering the objective, watch from the slide to avoid breaking the slide or damaging the objective. Screw the tube upwards until the object comes into focus.
3. If the condenser is movable by means of a spiral or rack and pinion screw, adjust its position below the stage, so that the source of light is in focus at the same time as the object. If the light comes from an electric lamp. Get the outline of the bulb in sharp focus, then adjust the mirror slightly so that full illumination is given.
4. Adjust the iris diaphragm by closing it gradually until the object appears well defined. At this stage. Use the fine adjustment to get the object in sharp focus.

PHARMACOGNOSY

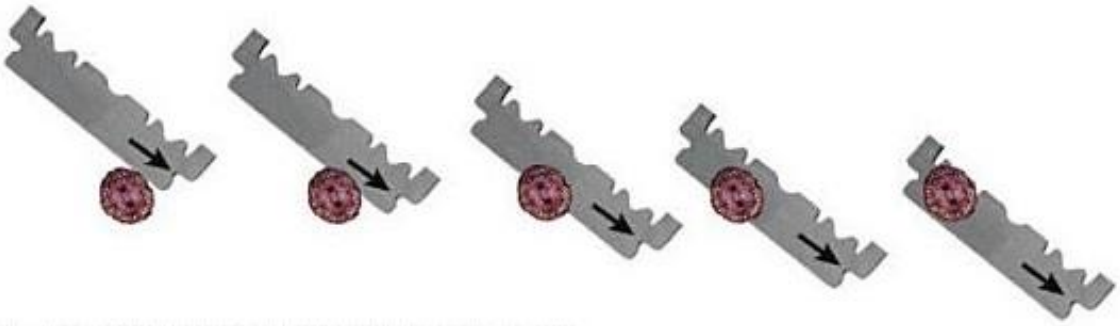
Obtaining of the critical illumination is very important for microscopic work. It can be achieved by microscopic possessing a movable condenser and an iris diaphragm, using the flat mirror.



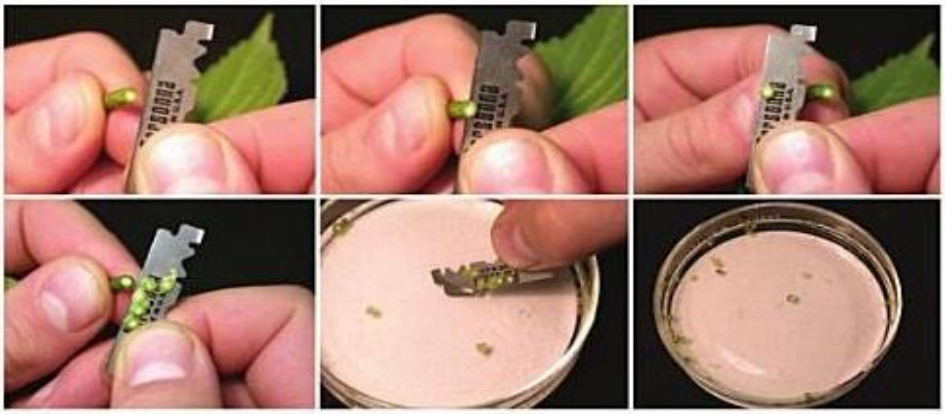




PHARMA







PHARMA



Method of sectioning small plant parts that are difficult to hold on their own. Styrofoam pieces can be used to support the tissue either by holding the tissue between two pieces or by making a small hole or slit in one Styrofoam piece to hold roots or leaves, respectively. Sections prepared in this way can be mounted directly in water under a cover glass or be stained before mounting.

Sectioning with support material

In cases where the plant tissue is not strong enough to be sectioned with the usual method, it can often be cut when supported by some other material. Leaves and petals usually require support. In the following example, a very small stem is sectioned with a Styrofoam support. If this is not available, some other material with the suitable consistency can be substituted.

Plant material testing: Method

1. Carefully using a pair of tweezers and a scalpel, slice a thin transverse section off the part of the plant. Many sections may be too thick but with practice some sections will be thin enough to use.
2. Place the plant sections in a petri dish of water to keep moist.
3. Repeat steps 2 and 3 to produce a number of sections from each plant part being investigated.
4. Remove plant section from the petri dish and place on a slide.
5. Remove excess water by carefully touching the edge of the section with absorbent paper.
6. Add 3 drops of Chloral hydrate to remove chlorophyll or any colouring matter in the section. Dry or remove excess solution with a absorbent paper.
7. Add 1 to 2 drops of stain to the section. Phloroglucinol should be left for 4 minutes before removing excess stain and adding a drop of HCl.
8. Remove excess stain with absorbent paper as before.
9. Slowly lower the cover slip onto the section using the mounted needle making sure that air bubbles are removed from the slide.
10. Label the slide and examine under the microscope.

Clearing

This technique is especially useful for examination of the intact vascular systems of leaves and floral parts. However, it may also be resorted to as a means of making thick freehand sections more transparent. The easiest method is to clear tissue by incubating it in EtOH to remove hydrophobic pigments including chlorophyll. This may take several hours.

Many such plant parts of either fresh, preserved or dry materials may be cleared sufficiently by warming them in chloral hydrate.

Others may require treatment with NaOH as follows:

Clear in 5% NaOH in a Petri dish in an oven (the time varies from one to several day depending on the material). Wash 3 to 5 times in distilled water carefully with a pipette. If more clearing appears to be necessary, use a saturated aqueous solution of chloral hydrate for 24 hours. Wash again in distilled water.

Clearing solution: **Chloral hydrate** is used to clear whole structures or sections which are not otherwise transparent. Mount the specimen in a few drops of **chloral hydrate.** Warm the mount over alcohol lamp until it seems more transparent. Do not allow the fluid to boil

PHARMACOGNOSY

violently. Observe the mount directly, or rinse and remount in glycerin. This treatment renders cell walls visible, but removes most cell contents. Starch grains are dissolved, but crystals of calcium oxalate are not. Lignin also remains and tests positive with **phloroglucinol**.

Staining

Two or three drops of stain can be transferred by pipette to a spot plate, ceramic ring slide, or depression slide. Transfer the specimen by forceps to the drop of dye and wait 1-several minutes (depending on the dye concentration). Transfer the specimen to a drop of solvent (water or dilute ethanol) of the same concentration used to make up the dye. Let stand a minute.

Alternatively, sections can be placed in gooch crucibles, and moved among small beakers, syracuse watch glasses, or coplin jars if production staining is necessary.

Staining solution: Add **Phloroglucinol** solution to the material to be stained. Let stand for 1 minute. Add 1 drop of **hydrochloric acid** solution. Leave for 5minutes. To enhance the reaction, the material can be air dried.

Lignin: Prepare a wet mount of sectioned or cleared material in phloroglucinol in HCL. Lignin will stain red.

Mounting

Take a clean dry microscope slide and place on it a small meniscus of calcium chloride solution. Transfer a stained specimen to the calcium chloride and wait until the specimen mixes well with the salt solution. Drop a clean cover slip onto the specimen.

The student should become familiar with the use of the following simple techniques and apply them in the study of laboratory materials:

Mounting solution: Dilute **glycerin** is used for preparing wet mounts for general observations when rapid drying of the mount is undesirable.
"Wet" water is used for preparing wet mounts of specimens for general observations with a minimum of trapped air.

BEWARE OF THE ACID. DON'T ALLOW ANY CHEMICALS TO COME IN CONTACT WITH THE MICROSCOPE AND ITS LENSES.

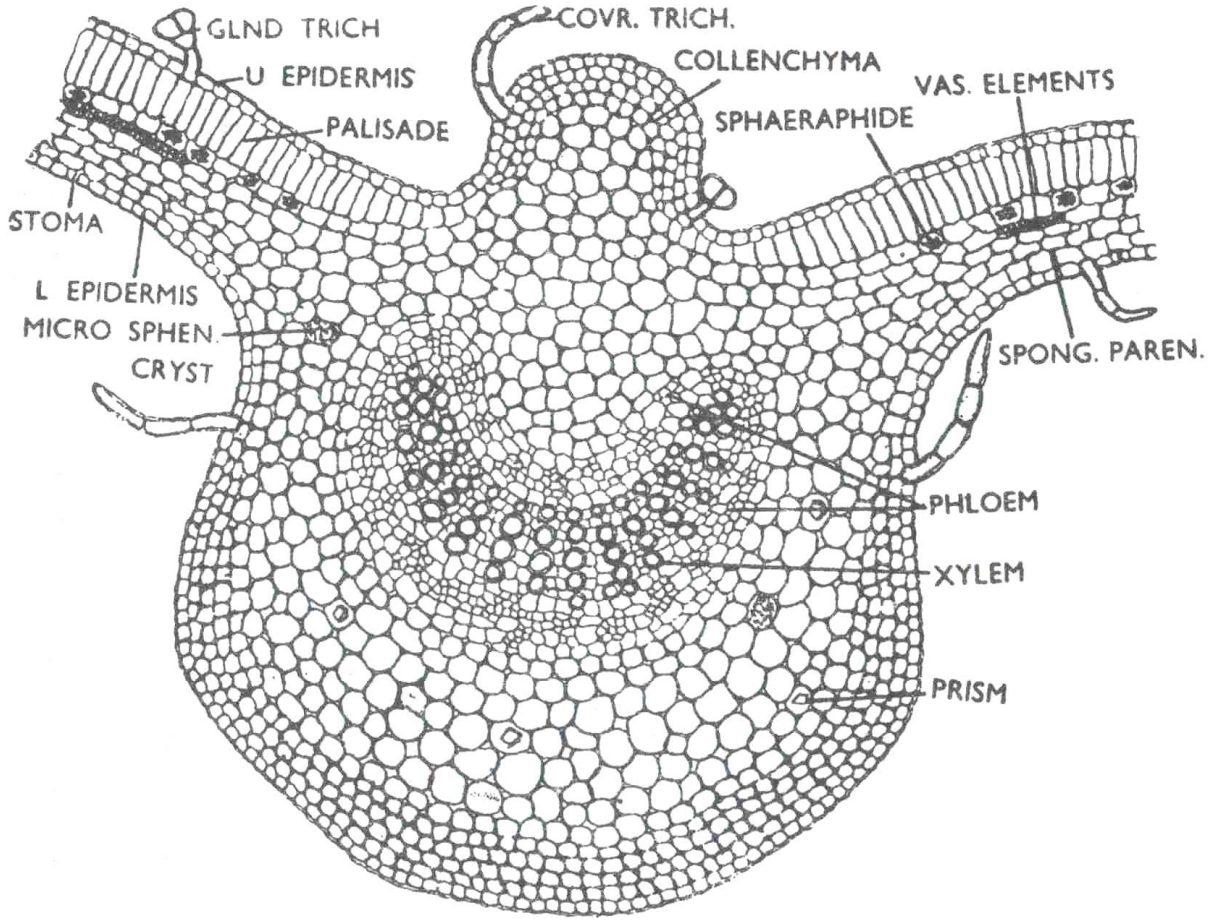
PHARMACOGNOSY

TRANSVERSE SECTION OF DATURA LEAF

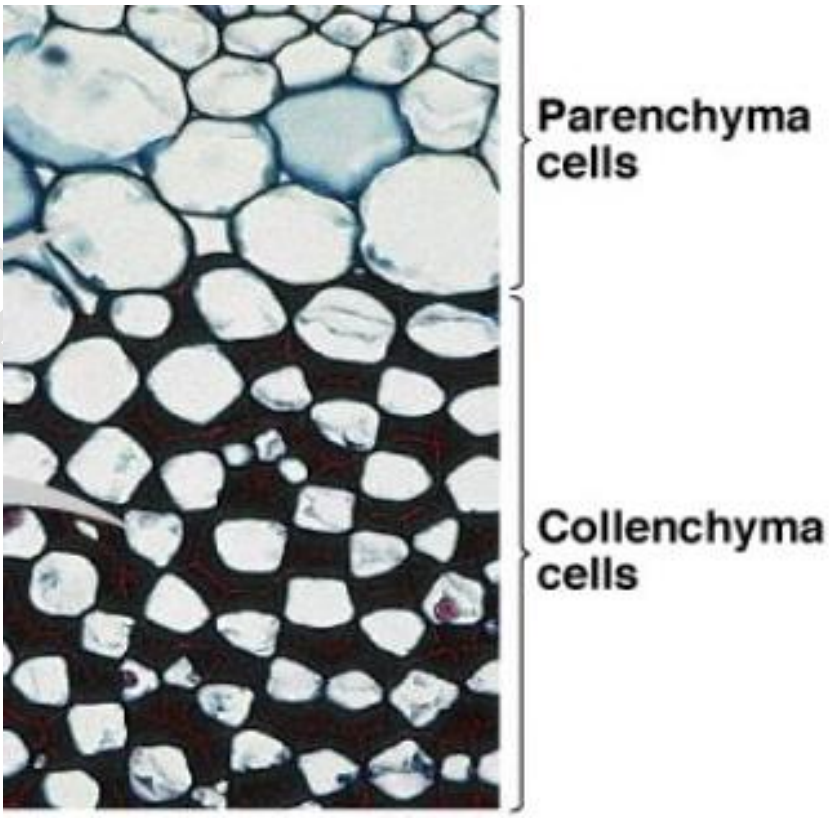
Microscopical characters

- A transverse section of a leaf shows that it has a bifacial structure.
- Both surfaces are covered with a smooth cuticle and possess both stomata and hairs.
- The following characters were observed in the **Lamina** and **Midrib** region**:**
- **Lamina**
  - In the lamina it has the upper epidermis which is single layer, rectangular cells covered with cuticle. Both covering and glandular trichomes are present. The covering trichomes are uni-seriate, multicellular, warty and with blunt apex. The glandular trichomes have one stalk consisting of one cell and multiheaded.
  - The mesophyll has spongy parenchyma and palisade parenchyma in it. Palisade cells are rapidly elongated, single layer and compactly arranged. Spongy parenchyma are several layers, loosely arranged consisting of micro-spenoidal crystals and vascular strands.
  - Cluster crystals of calcium oxalate are abundant in the mesophyll and microsphenoidal and prismatic crystals are also found.
  - The stomata are of the anisocytic and anomocytic types.
  - The epidermal cells have wavy walls, particularly those of the lower epidermis.
  - If portions of the leaf are cleared with chloral hydrate solution, the abundance of the cluster crystals of calcium oxalate and their distribution with regard to the veins may be noted.
- **Midrid**
  - The midrib shows a bicollateral structure and characteristic subepidermal masses of collenchyma appear below the upper and above the lower epidermis followed by the cortical parenchymatous cells containing calcium oxalate crystals.
  - The lower epidermis is similar to that of the upper epidermis but has more number of trichomes and stomata when compared with upper epidermis.
  - The xylem forms a strongly curved arc.
  - Sclerenchyma is absent.

PHARMACOGNOSY



Datura Leaf. T. S. (X 50)



Transverse section of clove

Microscopic characters of clove:

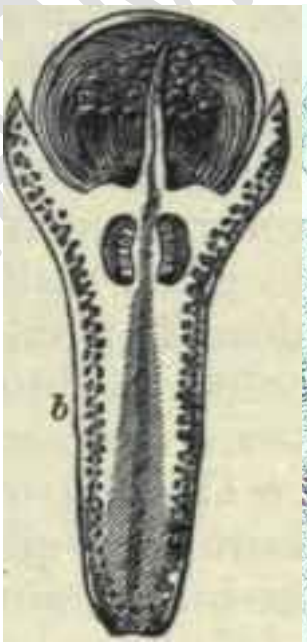
A transverse section through hypanthium shows:

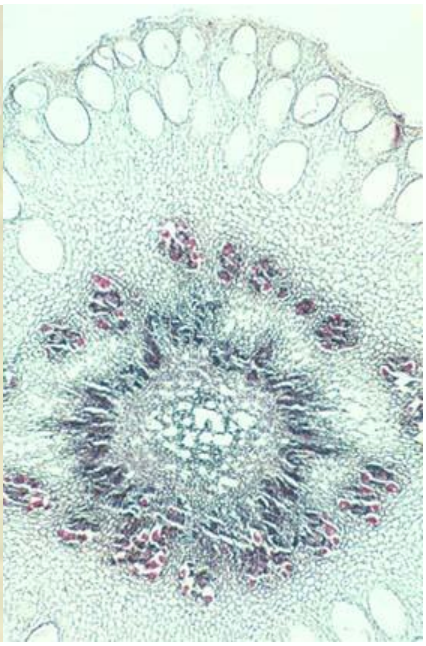
1. Epidermis with thick cuticle and stomata.
2. Collenchymatous parenchyma containing numerous schizolysigenous oil glands.
3. Calcium oxalate clusters in parenchyma.
4. Zone of slightly thick walled cells embedding a ring of about 15 bilateral vascular bundles.
5. Meristele enclosed by lignified pericyclic fibres.
6. Xylem of 5-6 lignified vessels.
7. Parenchyma containing air spaces or lacuna (aerenchyma).
8. Central columella:

   Thick walled parenchymatous cells containing cluster crystals of calcium oxalate and 20-25 small vascular bundles.

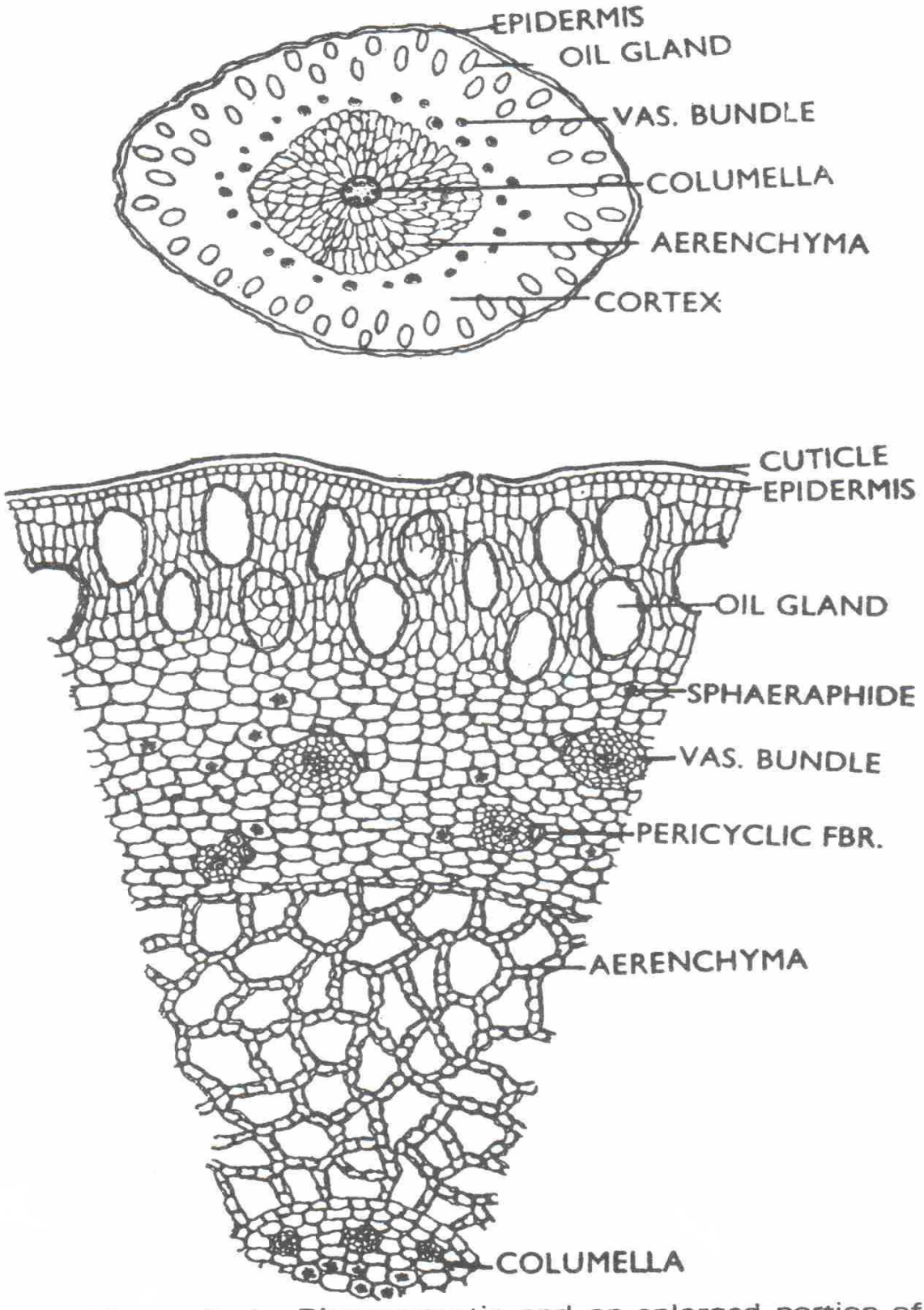
A transverse section through region of ovary shows:

1. Ovary with ovarian wall.
2. Parenchymatous dessepiment.
3. Ovules with axile placentation.
4. Starch, prisms of calcium oxalate and stone cells are absent.





PHARMACOGNOSY

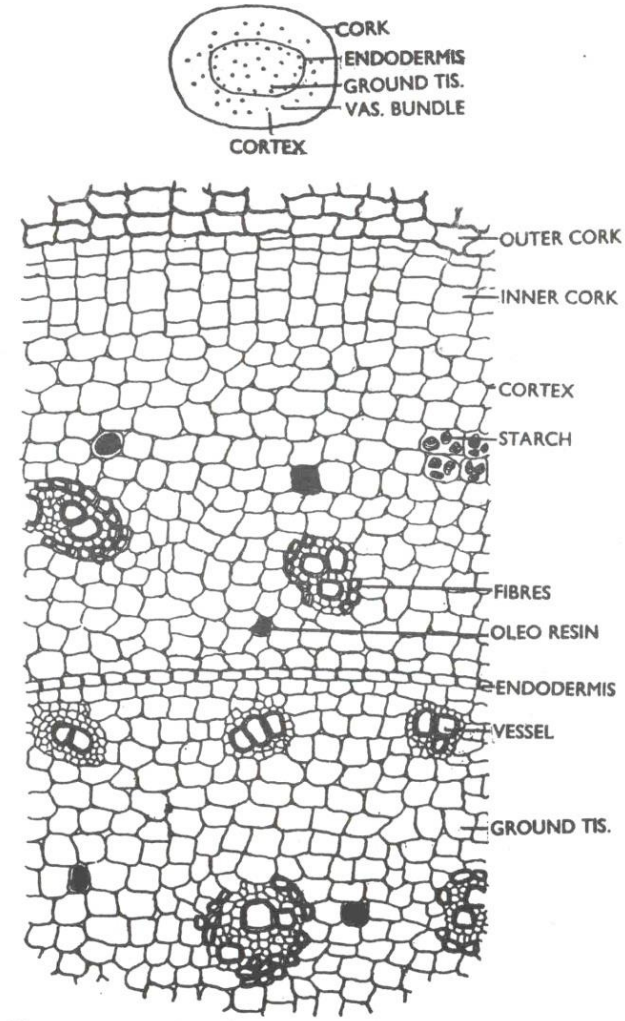


Clove Flower Bud. Diagrammatic and an enlarged portion of T. S. (× 50)

Transverse section of ginger

A transverse section of ginger shows:

1. **Cork**: outer zone of irregularly arranged cells and inner or radially arranged cells (absent in Jamaican ginger).
2. **Cortex**: Thin walled, cellulosic, rounded parenchyma with intercellular spaces.
3. **Starch** grains in parenchyma of cortex.
4. Closed collateral fibro-vascular bundles in the cortex.
5. Brown oleoresin cells.
6. **Stele**: a ring of vascular bundles (without fibres) just below the endodermis, ground tissue of parenchymatous cells with fibro vascular bundles, oleoresin cells and starch.
7. **Xylem vessels**: annular, spiral or reticulate thickenings unlignified.
8. **Fibres**: thin walled with only central lumen lignified with pectosic transverse septa.



Ginger Rhizome. Diagrammatic and an enlarged portion of T. S. (X 100)

PHARMACOGNOSY